



# LR0.FM: LOW-RESOLUTION ZERO-SHOT CLASSIFICATION

## BENCHMARK FOR FOUNDATION MODELS

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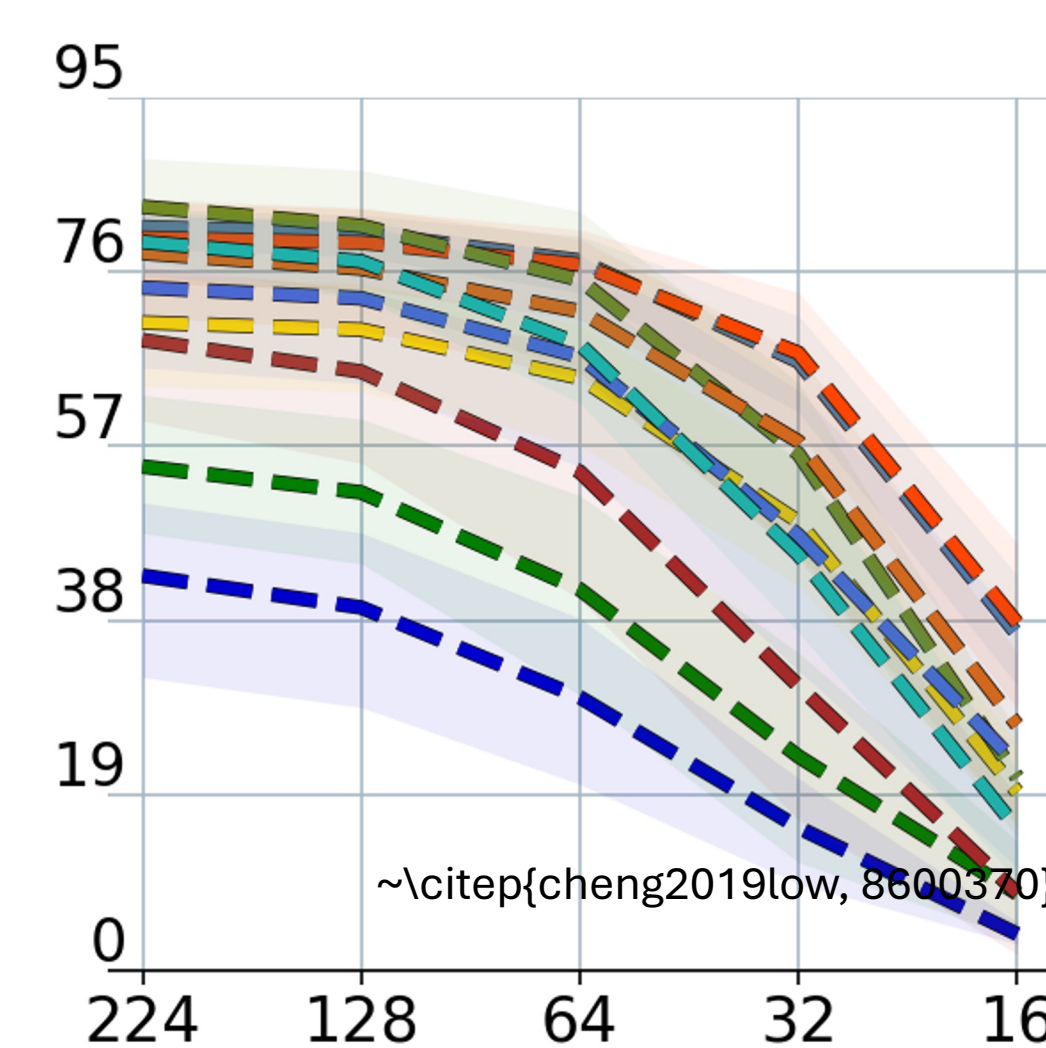


## Introduction

- Visual-language models (FM) not robust against low-resolution.
- **LR0.FM** benchmarks impact of LR on zero-shot classification of 10 FM (66 backbones & 15 datasets).
- **Weighted Aggregated Robustness (WAR)** improves over existing evaluation metric.
- **LR-TK0** enhances robustness without compromising pre-trained weights.

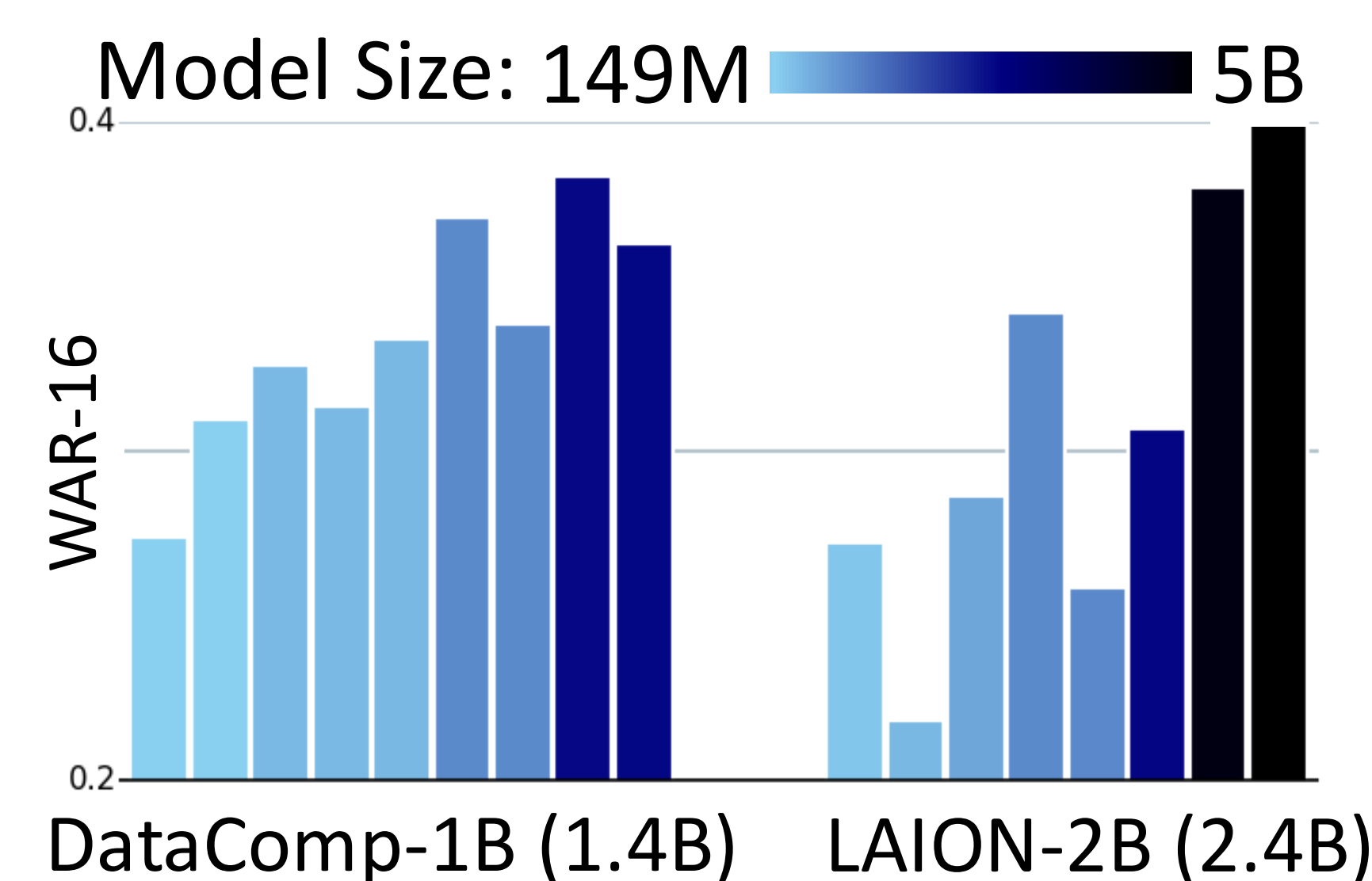


Maraca (toy)

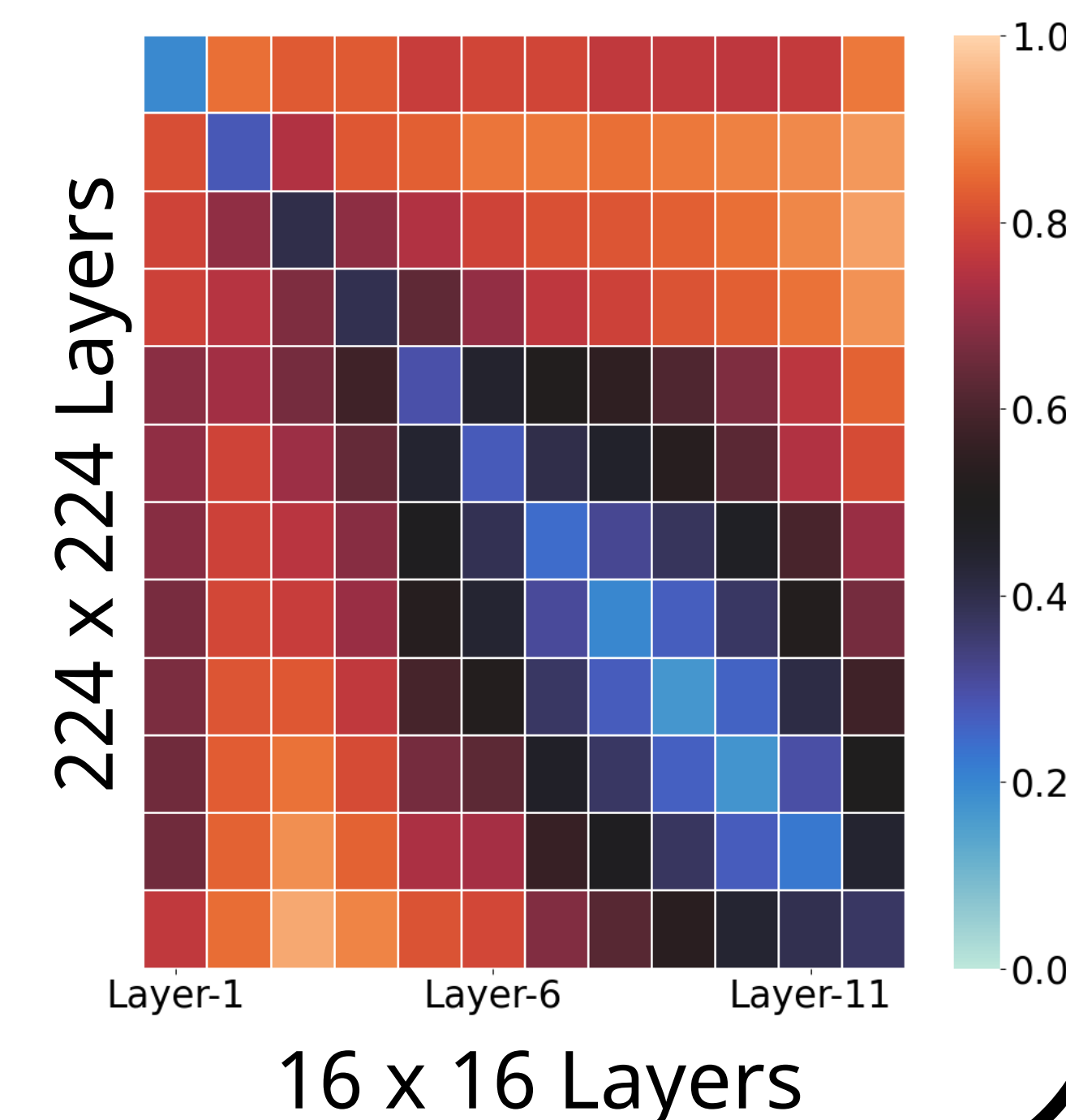


## Benchmark : LR0.FM

- Pretraining Size not important



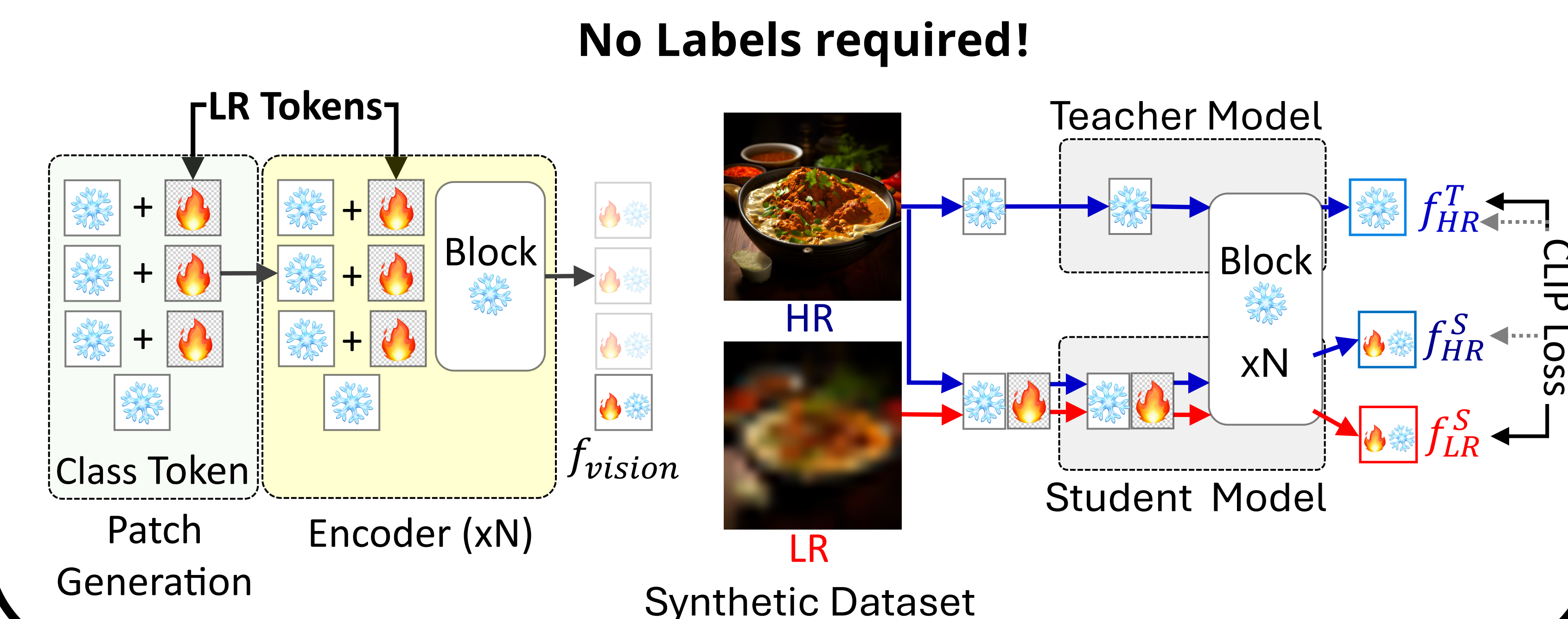
- Initial layers more dissimilar than the deeper layer



## LR-TK0

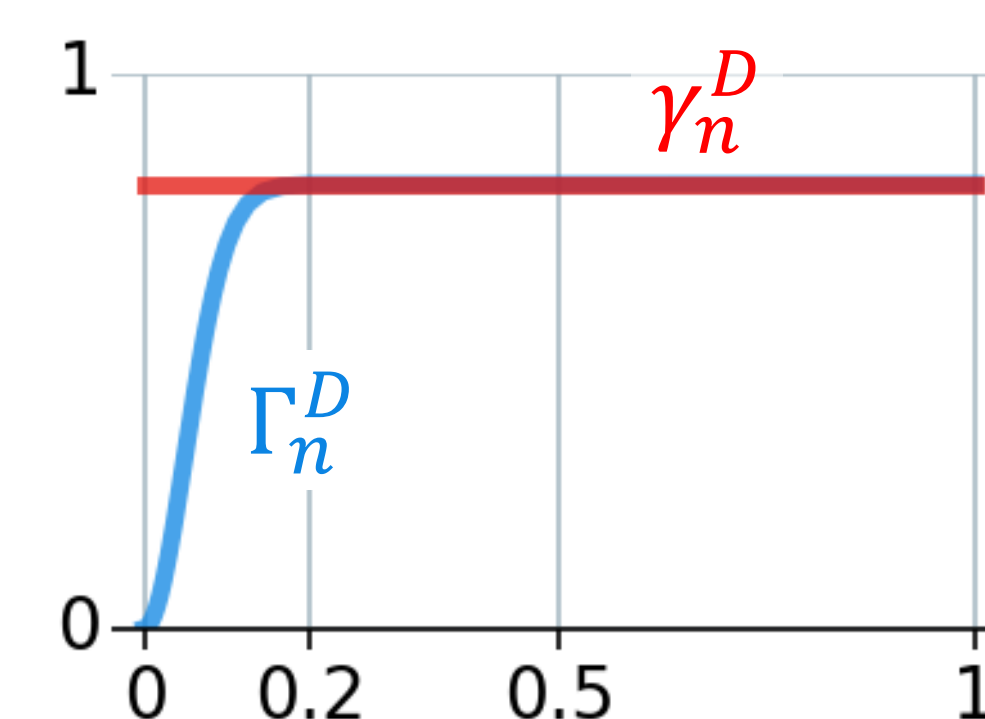
Spatial & Class Tokens & Frozen Parameters

Trainable LR Tokens

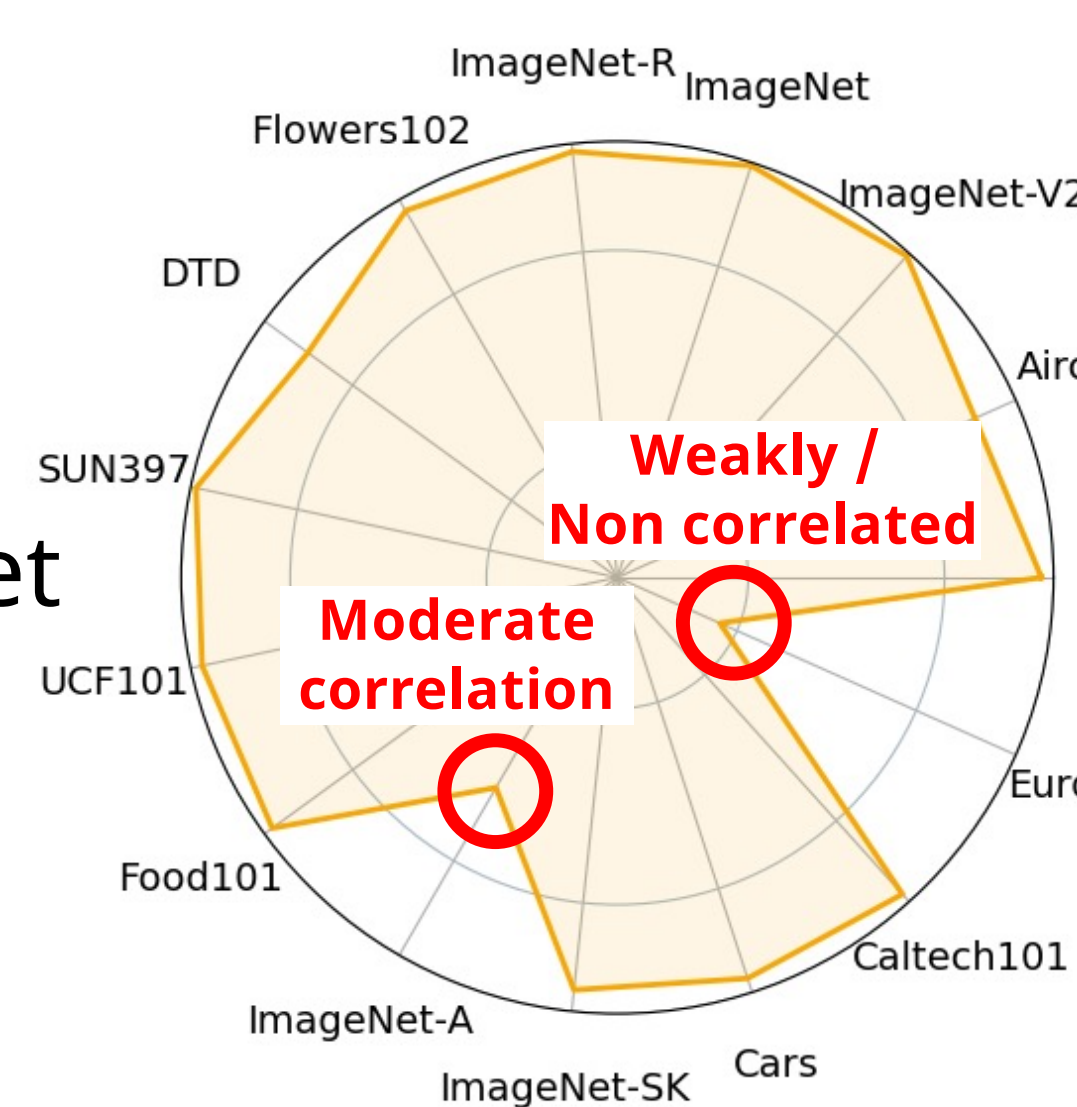


## Weighted Aggregated Robustness

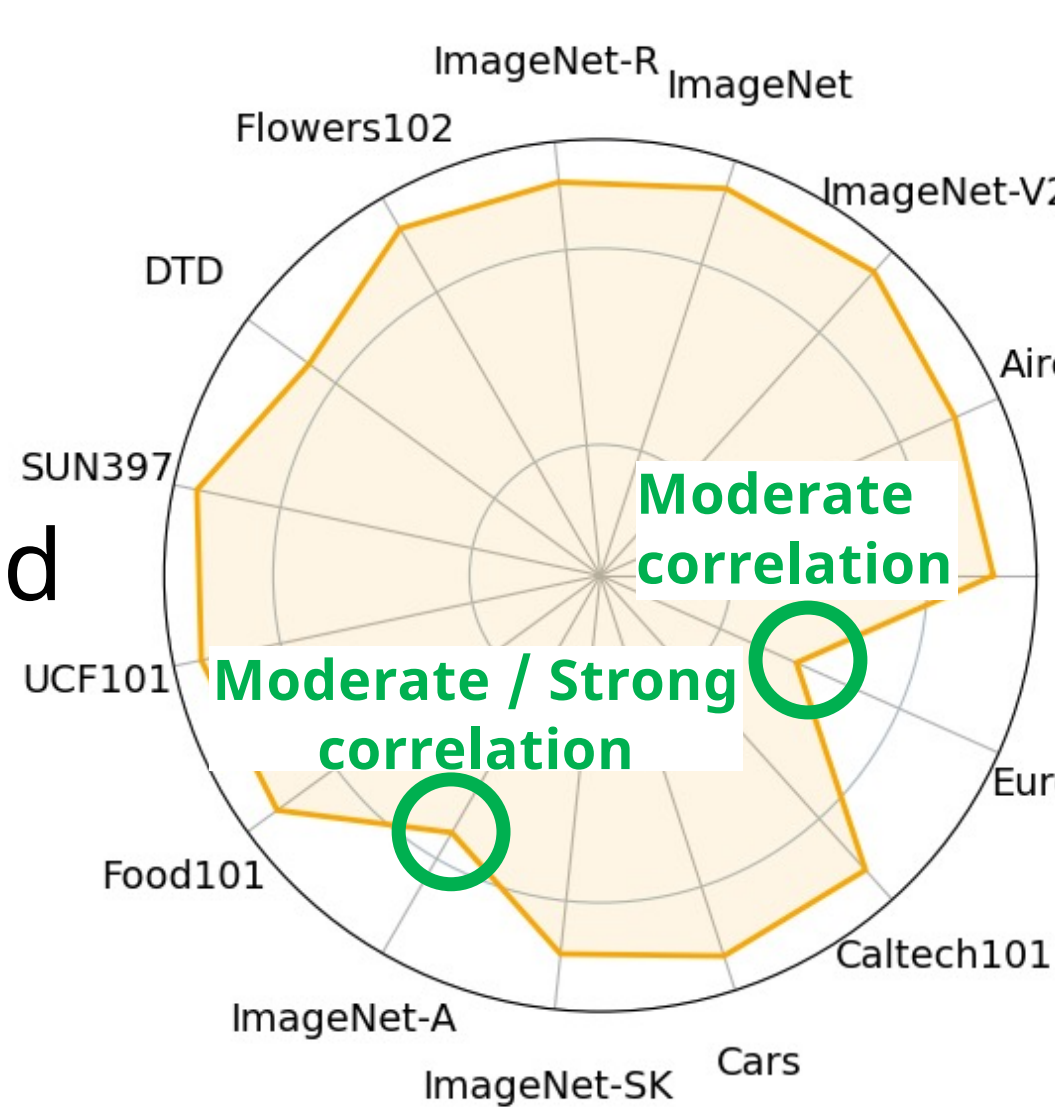
Proposed improved robustness  $\Gamma_n^D = \gamma_b^d \times (1 - e^{-\alpha \epsilon^2})$   
Traditional robustness  $\gamma_b^d$



Uniform Weight for each dataset ignores dataset

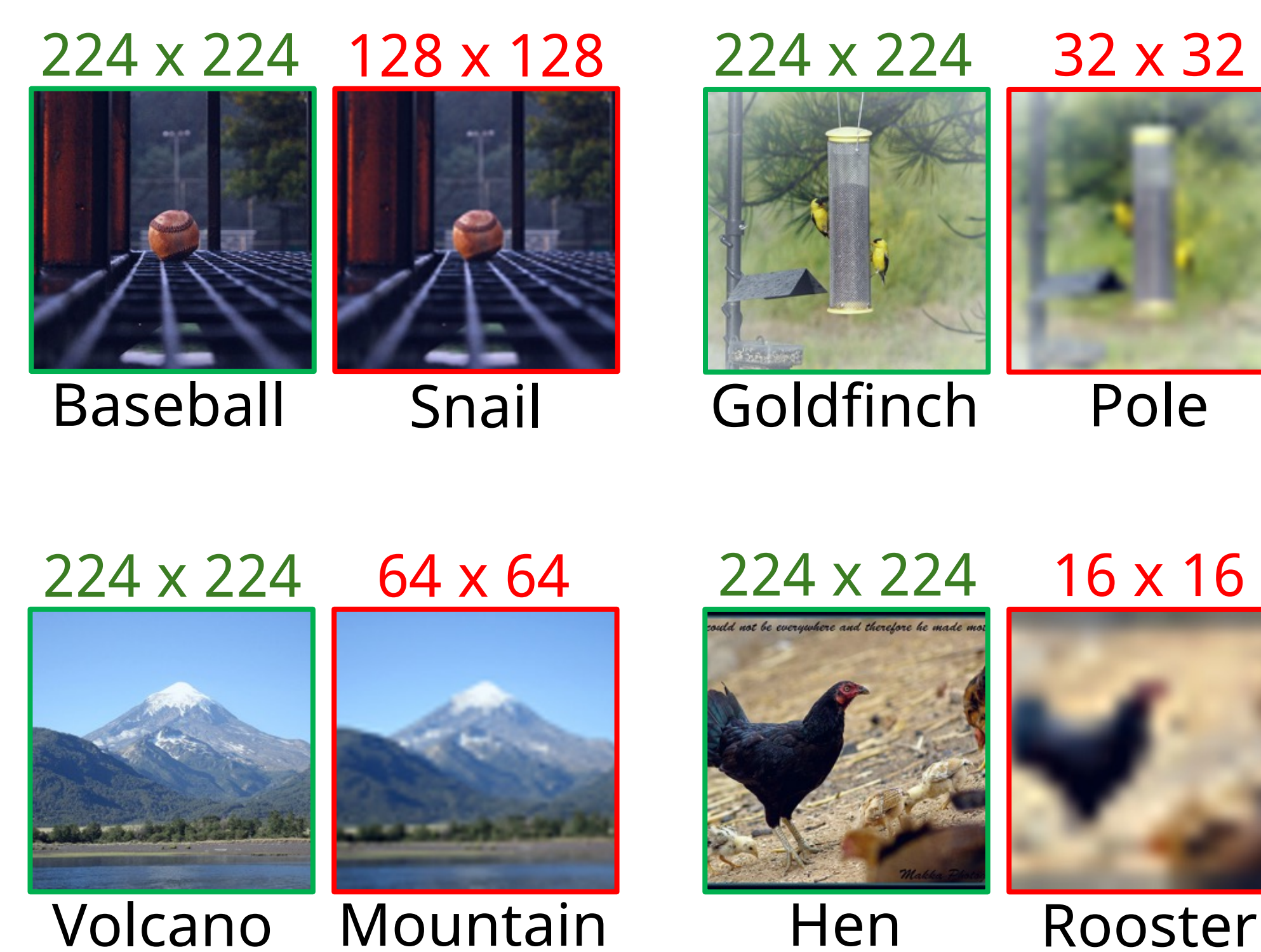


WAR: Weighted average



## Towards Solution

- Model makes correct predictions in low-resolution
- Preserve pretrained weights
- Zero shot evaluation (Diffusion generated training images)



"wedding under the oak trees"



## Results

